LangSmith Traceability, Recall, Precision & GenAI Evaluation Metrics (Answer Key)

Q1. What is the primary purpose of traceability in LangSmith?

A. To increase model training speed

B. To track and analyze the execution flow of LLM applications

C. To compress embeddings

D. To reduce API costs

Answer: B

Q2. In LangSmith, a trace typically represents:

A. A database backup

B. A network connection

C. A record of an application's execution and interactions

D. A model checkpoint

Answer: C

Q3. Recall measures:

A. How many retrieved results are relevant

B. How many relevant results were successfully retrieved

C. The speed of retrieval

D. The size of the knowledge base

Answer: B

Q4. A retrieval system found 8 relevant documents out of 10 total relevant documents available. What is its Recall?

A. 80%

B. 20%

C. 100%

D. 50%

Answer: A

Q5. Precision measures:

A. The percentage of retrieved items that are relevant

B. The percentage of all relevant items retrieved

C. The latency of the system

D. The token count of responses

Answer: A

Q6. A search system retrieves 20 documents, but only 15 are relevant. What is the Precision?

A. 50%

B. 75%

C. 25%

D. 90%

Answer: B

Q7. Which evaluation metric is commonly used to determine whether an LLM response is factually correct?

A. Correctness

B. Temperature

C. Token Usage

D. Context Length

Answer: A

Q8. In a RAG application, low Recall usually indicates that:

A. Too many irrelevant documents are retrieved

B. Relevant information is being missed during retrieval

C. The model is generating faster responses

D. The embedding size is too large

Answer: B

Q9. Which LangSmith feature helps developers identify where an LLM application failed during execution?

A. Prompt Templates

B. Traces and Runs Visualization

C. Vector Databases

D. Fine-Tuning

Answer: B

Q10. Why are evaluation metrics important in Generative AI systems?

A. They increase GPU memory

B. They help measure and improve model quality and reliability

C. They automatically train the model

D. They eliminate hallucinations completely

Answer: B